

**Question:**

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

**Reasoning + Answer:**

<think>

Given that the automobile battery produces a voltage of 12 volts , we want to find the current and resistance of a head light bulb that dissipates 40 watts of power when connected directly across the battery terminals . We know that power ( P ) is equal to voltage ( V ) multiplied by current ( I ), or  $P = V I$ .

We can rearrange this to solve for current:  $I = P / V$ .

We also know that power ( P ) is equal to the product of the voltage ( V ) and resistance ( R ), or  $P = V R$ .

Rearranging for resistance gives us  $R = P / V$ .

Now , we can substitute the given values into these formulas:  $I = 40 \text{ watts} / 12 \text{ volts}$  and  $R = 40 \text{ watts} / 12 \text{ volts}$ .

</think><answer> E.